Chapter 12. Quotients

Cosets, quotient groups and spaces, isomorphism theorems

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Quotient groups and quotient spaces raise our language one level and supply powerful abstract tools; in particular, induction on dimension via quotient spaces becomes a basic technique of linear algebra.

12.1 Cosets

Motivation. The left cosets xH of a subgroup partition the group into pieces of equal size. Counting them gives Lagrange's theorem: the order of a subgroup divides the order of the group, which already classifies groups of prime order.

Definition 12.1.1 (Cosets)

For $H \leq G$, $x \in G$: $xH = \{xh : h \in H\}$ (or $x + H$ additively) is the **left coset** of H with **representative** x, and Hx the **right coset**. "Coset" means left coset from now on.

Observation 12.1.2

For $H \leq G$, $x, y \in G$: (a) $x \in xH$, so $G = \bigcup_{x \in G} xH$;
(b) $xH = yH$ iff $y^{-1}x \in H$;
(c) $xH \cap yH = \emptyset$ if $xH \neq yH$.

Proof. (a) $x = x1$. (b) If $xH = yH$, then $x \in yH$ by (a), $x = yh'$, so $y^{-1}x = h' \in H$. Conversely, if $y^{-1}x = h_0 \in H$, then $xh = yh_0h \in yH$, so $xH \subseteq yH$; likewise $yH \subseteq xH$ (as $x^{-1}y = h_0^{-1} \in H$). (c) If $xh_1 = yh_2$, then $y^{-1}x = h_2h_1^{-1} \in H$, so $xH = yH$ by (b). ■

So the distinct cosets partition G, which is the equivalence relation

$$x \equiv y \pmod{H} \iff xH = yH \iff y^{-1}x \in H,$$

with classes $\bar{x} = xH$. The set of cosets $G/H = \frac{G}{H} = \{xH : x \in G\} = \{\bar{x} : x \in G\}$ is the **coset space** of G modulo H. For a **complete set of representatives** $\{x_i\}_{i \in I}$, $G/H = \{\bar{x}_i\}$ and $G = \coprod_i x_iH$ (**coset decomposition**).

Example 12.1.3

$1H = H$ is always a coset.

Exercise 12.1.4

$|G/G| = 1$ and $|G/\{1\}| = |G|$.

Solution. $xG = G$ for all x; $x\{1\} = \{x\}$ are all distinct. □

Example 12.1.5

The solution set $X_0 + \ker L_A$ of $AX = B$ is a coset of the subgroup $\ker L_A$ of the abelian group $(F^n, +)$.

Example 12.1.6

For $n\mathbb{Z} \leq \mathbb{Z}$ ($n \neq 0$): $\bar{a} = a + n\mathbb{Z} = \bar{b} \iff a - b \in n\mathbb{Z} \iff n \mid (a-b)$, i.e. a and b have the same remainder mod n. So $\mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, \ldots, \overline{n-1}\}$, the residues mod n. In number theory $\bar{a}$ is the **residue class** of a, $a \equiv b \pmod{n}$, and a complete set of representatives is a **complete residue system**.

Example 12.1.7

In S_3 (notation of Example 11.8.4) with $H = \{1, \sigma\}$: $\tau H = \{\tau, \sigma\tau^2\}$, $\tau^2 H = \{\tau^2, \sigma\tau\}$, so $S_3/H = \{\bar{1}, \bar{\tau}, \overline{\tau^2}\}$. Since $\sigma\tau \neq \tau\sigma$, $\tau H \neq H\tau$: left and right cosets need not coincide.

Exercise 12.1.8

For $K = \{1, \tau, \tau^2\} \leq S_3$: $|S_3/K| = 2$, and left and right cosets coincide (e.g. $\sigma K = K\sigma$).

Solution. $S_3 = K \amalg \sigma K$ with $\sigma K = \{\sigma, \sigma\tau, \sigma\tau^2\}$, the three transpositions. Since $K\sigma$ is a coset of size 3 disjoint from K, it equals σK. □

Observation 12.1.9

For $H \leq G$, $x, y \in G$: $|xH| = |yH| = |H|$.

Proof. $\lambda_x : H \to xH$, $h \mapsto xh$, is a bijection (injective by cancellation, onto by definition). ■

Observation 12.1.10 (Lagrange's theorem)

If G is finite and $H \leq G$, then $|H|$ divides $|G|$ and $|G/H| = \frac{|G|}{|H|}$.

Proof. The distinct cosets partition G and each has $|H|$ elements (Observation 12.1.9). ■

Notation 12.1.11

If $|G/H| < \infty$ (G may be infinite), $[G : H] = |G/H|$ is the **index** of H in G.

Observation 12.1.12

If $K \leq H \leq G$ with $|G/H|, |H/K| < \infty$, then $[G : K] = [G : H][H : K]$.

Proof. Let $\{g_i\}$ be a complete set of representatives for G/H and $\{h_j\}$ for H/K. Then $\{g_i h_j\}$ is one for G/K: $G = \coprod_i g_i H = \coprod_i g_i \coprod_j h_j K = \bigcup_{i,j} g_i h_j K$, and the union is disjoint, since $g_i h_j K = g_{i'} h_{j'} K$ forces $g_i H = g_{i'} H$ (both cosets lie in it), so $i = i'$, and then $h_j K = h_{j'} K$, so $j = j'$. ■

Example 12.1.13

By Example 11.3.6(e) and Lagrange, $\mu_m \leq \mu_n$ iff $m \mid n$.

Observation 12.1.14

If G is finite and $x \in G$, then $|x|$ divides $|G|$.

Proof. $|x| = |\langle x \rangle|$; apply Lagrange. ■

Proposition 12.1.15

If $|G| = p$ is prime, then G is cyclic, $G \approx \mu_p$.

Proof. Take $x \neq 1$. Then $|x| \neq 1$ divides p, so $|x| = p$, $\langle x \rangle = G$, and $\langle x \rangle \approx \mu_p$ (Exercise 11.7.9). ■

Example 12.1.16

Subgroups of S_3: besides $\{id\}$ and S_3, Lagrange allows orders $2, 3$, and such subgroups are cyclic (Proposition 12.1.15). Elements of order 2 give $\{id, (1\,2)\}$, $\{id, (2\,3)\}$, $\{id, (1\,3)\}$; elements of order 3, $(1\,2\,3)$ and $(1\,3\,2) = (1\,2\,3)^2$, give the unique subgroup $\{id, (1\,2\,3), (1\,3\,2)\}$ of order 3.

Exercise 12.1.17

If $|G|$ is even, G has an element of order 2.

Solution. Pair each x with x^{-1}. The pairs with $x \neq x^{-1}$ have 2 elements, so the set $\{x : x = x^{-1}\}$ has even size; it contains 1, hence some $x \neq 1$ with $x = x^{-1}$, i.e. $|x| = 2$. □

Remark 12.1.18

But if $d \mid |G|$, G need not have an element, or even a subgroup, of order d.

Exercise 12.1.19

(a) S_4 and A_4 have no elements of order 6, 8 or 12. (b) S_4 has exactly four subgroups of order 6. (c) A_4 has no subgroup of order 6. (d) How many subgroups of order 8 does S_4 have? (e) Of order 12?

Solution. (a) Elements of S_4 have cycle types $1, (2), (3), (4), (2,2)$, of orders $1, 2, 3, 4, 2$.

(b) A subgroup of order 6 fixing a point i is the copy of S_3 on the other three points; there are four of these. Conversely, a subgroup H of order 6 has an element of order 3 (Cauchy, Theorem 12.1.20), a 3-cycle fixing some i, and the order-2 elements of H normalize $\langle 3\text{-cycle} \rangle$, which forces them to fix i too, so H is the stabilizer of i.

(c) A_4 (order 12) has eight 3-cycles and three elements of type $(2,2)$. A subgroup H of order 6 would have index 2, hence be normal (Observation 12.2.9) and contain all squares, in particular all eight 3-cycles ($c = (c^2)^2$), which is too many.

(d) Three (the dihedral groups D_4, Sylow 2-subgroups).

(e) One, namely A_4 (a subgroup of index 2 contains all squares, which generate A_4). □

Theorem 12.1.20 (Cauchy)

If a prime p divides $|G|$, then G has an element of order p.

Proof [Supplement] (McKay; the book defers it to abstract algebra). Let $X = \{(x_1, \ldots, x_p) \in G^p : x_1 x_2 \cdots x_p = 1\}$. Choosing $x_1, \ldots, x_{p-1}$ freely determines x_p, so $|X| = |G|^{p-1}$, divisible by p. If $(x_1, \ldots, x_p) \in X$, then $x_2 \cdots x_p x_1 = x_1^{-1}(x_1 \cdots x_p)x_1 = 1$, so cyclic

shifts preserve X. The orbits of the cyclic shift have size 1 or p (p prime), and orbits of size 1 are the constant tuples $(x, \dots, x)$ with $x^p = 1$. Since $|X| \equiv 0 \pmod p$ and the size-p orbits contribute multiples of p, the number of x with $x^p = 1$ is divisible by p. It is at least 1 ($x = 1$), hence at least $p \geq 2$. So some $x \neq 1$ has $x^p = 1$, i.e. $|x| = p$. ■

12.2 Normal subgroups and quotient groups

Motivation. Multiplying cosets by representatives, $\bar{x}\,\bar{y} = \overline{xy}$, is well defined exactly when the subgroup is normal. Then G/N is a group, the natural projection is a homomorphism with kernel N, and residue arithmetic mod n becomes the group $\mathbb{Z}_n$.

Residue arithmetic ($3^3 - 20 \cdot 3 + 9 \equiv 3^3 + 0 + 4 \equiv 1 \pmod 5$), like adding fractions with different representatives, raises the question of *well-definedness*. To make G/H a group, the natural operation is $(xH)(yH) = (xy)H$, i.e. $\bar{x}\,\bar{y} = \overline{xy}$; we must check that if $\bar{x} = \bar{x}'$ and $\bar{y} = \bar{y}'$, then $\overline{xy} = \overline{x'y'}$, i.e. $x'^{-1}x, y'^{-1}y \in H \Rightarrow (x'y')^{-1}xy = y'^{-1}x'^{-1}xy \in H$.

Example 12.2.1

For $H = \{1, \sigma\} \leq S_3$ (Example 12.1.7): $\bar{1} = \bar{\sigma}$ and $\bar{\tau} = \overline{\sigma\tau^2}$, but $1 \cdot \tau = \tau$ while $\sigma \cdot \sigma\tau^2 = \tau^2$, and $\bar{\tau} \neq \overline{\tau^2}$. So the operation is not well defined. (For $K = \{1, \tau, \tau^2\}$ it is.)

Definition 12.2.2 (Normal subgroup)

$H \leq G$ is **normal**, $H \trianglelefteq G$, if $g^{-1}hg \in H$ for all $g \in G$, $h \in H$.

Observation 12.2.3

For $H \leq G$, $\bar{x}\,\bar{y} = \overline{xy}$ is well defined on G/H iff $H \trianglelefteq G$.

Proof. If $H \trianglelefteq G$ and $x'^{-1}x, y'^{-1}y \in H$, then $y'^{-1}x'^{-1}xy = (y'^{-1}y)(y^{-1}(x'^{-1}x)y) \in H$. Conversely, if the operation is well defined, take $x = h$, $x' = 1$, $y = y' = g$: then $\bar{h} = \bar{1}$, so $\overline{hg} = \bar{g}$, i.e. $g^{-1}hg \in H$. ■

Proposition 12.2.4

If $N \trianglelefteq G$, then G/N with $\bar{x}\,\bar{y} = \overline{xy}$ is a group, the **quotient group** of G modulo N.

Proof. $(\bar{x}\bar{y})\bar{z} = \overline{xyz} = \bar{x}(\bar{y}\bar{z})$; $\bar{1}$ is the identity; $\bar{x}\,\overline{x^{-1}} = \bar{1} = \overline{x^{-1}}\,\bar{x}$, so $\bar{x}^{-1} = \overline{x^{-1}}$. ■

For $N \trianglelefteq G$, $n \in N$, $g \in G$: $ng = gn'$ with $n' = g^{-1}ng \in N$. So elements of N can be moved past elements of G, at the cost of changing n to n'.

Observation 12.2.5

For $H \leq G$ the following are equivalent: (1) $H \trianglelefteq G$, i.e. $g^{-1}Hg \leq H$ for all g; (2) $gHg^{-1} = H$ for all g; (3) $g^{-1}Hg = H$ for all g.

Proof. (3)⇔(2): replace g by g^{-1}. (3)⇒(1) is clear. (1)⇒(3): (1) for g^{-1} gives $gHg^{-1} \leq H$, i.e. $H \leq g^{-1}Hg$. ■

Exercise 12.2.6

For $H \leq G$ the following are equivalent: (1) $H \trianglelefteq G$; (2) $\varphi(H) = H$ for every inner automorphism φ; (3) $gH = Hg$ for all g.

Solution. (1)⇔(2) is Observation 12.2.5 with $\varphi = \mathrm{Int}(g)$. (3)⇔ $gHg^{-1} = H$ (multiply by g^{-1} on the right). □

Example 12.2.7

$\{1\} \trianglelefteq G$ and $G \trianglelefteq G$; $G/\{1\} \approx G$ and $G/G \approx \{1\}$.

Example 12.2.8

(a) In a commutative group every subgroup is normal, and G/N is commutative.
(b) If $G = \langle x \rangle$ is cyclic, so is $G/N = \langle \bar{x} \rangle$.

Observation 12.2.9

If $[G : H] = 2$, then $H \trianglelefteq G$.

Proof. $G = H \amalg xH$ for any $x \notin H$. Let $g \in G$, $h \in H$; we show $ghg^{-1} \in H$. If $g \in H$, this is clear. Otherwise $g = xh_1$, and $ghg^{-1} = xh_2x^{-1}$ with $h_2 = h_1hh_1^{-1} \in H$. If $xh_2x^{-1} \notin H$, then $xh_2x^{-1} = xh_3$ for some $h_3 \in H$, so $h_2x^{-1} = h_3$ and $x = h_3^{-1}h_2 \in H$, a contradiction. ■

(For index $3, 4, \ldots$ this fails, e.g. Example 12.2.1.)

Example 12.2.10

Groups of order 6. By Cauchy there are x, y with $|x| = 2$, $|y| = 3$; $N = \langle y \rangle$ has index 2, so $N \trianglelefteq G$ and $G = N \amalg xN = \{1, y, y^2, x, xy, xy^2\}$. Then $x^{-1}yx \in N$ has order 3, so it is y or y^2. If $x^{-1}yx = y$, then $xy = yx$ and $|xy| = 6$, so $G \approx \mu_6$. If $x^{-1}yx = y^2$, then $G \approx S_3$ (Examples 11.8.4, 11.9.23). So up to isomorphism the groups of order 6 are μ_6 and S_3.

Observation 12.2.11

If $\varphi : G \to H$ is a homomorphism, then $\ker \varphi \trianglelefteq G$.

Proof [Supplement]. For $k \in \ker \varphi$, $\varphi(g^{-1}kg) = \varphi(g)^{-1}\varphi(g) = e$. ■

(But $\mathrm{im}\, \varphi$ need not be normal in H.)

Definition 12.2.12 (Natural projection)

For $N \trianglelefteq G$, $\pi : G \to G/N$, $\pi(x) = \bar{x}$, is the **natural projection**.

Exercise 12.2.13

(a) π is an epimorphism.
(b) $\ker \pi = N$.
(c) For the natural embedding $\iota : N \to G$, $\pi \circ \iota = \mathbf{1}$.

Solution. (a) $\pi(xy) = \overline{xy} = \bar{x}\bar{y}$, onto by definition.

(b) $\bar{x} = \bar{1} \iff x \in N$.

(c) is (b). □

The quotient operation is the unique one making π a homomorphism, so π carries all information about G/N. Residue calculus is the operation of $\mathbb{Z}/n\mathbb{Z}$.

Notation 12.2.14

For $1 \leq n \in \mathbb{Z}$, $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$, with $\bar{a} + \bar{b} = \overline{a+b}$.

Observation 12.2.15

$\mathbb{Z}_n$ is cyclic, so $\mathbb{Z}_n \approx \mu_n$.

Proof. $\mathbb{Z}_n = \langle \bar{1} \rangle$ has order n; apply Exercise 11.7.9. ■

Exercise 12.2.16

(a) In A_4, $N = \{id, (1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3)\}$ satisfies $N \approx \mathbb{Z}_2 \times \mathbb{Z}_2$ and $N \trianglelefteq A_4$.
(b) $H = \{id, (1\,2)(3\,4)\}$ satisfies $H \trianglelefteq N$ and $N \trianglelefteq A_4$, but H is not normal in A_4.

Solution. (a) Each non-identity element of N has order 2 and the product of two of them is the third, so N is a subgroup in which every element squares to 1: the Klein 4-group. Conjugation preserves cycle type, and N consists of the identity and all elements of type $(2,2)$, so N is normal (even in S_4).

(b) H has index 2 in N, so it is normal in N. But $(1\,2\,3)^{-1}(1\,2)(3\,4)(1\,2\,3) = (1\,4)(2\,3) \notin H$ (a normal subgroup of a normal subgroup need not be normal). □

Remark 12.2.17

The normal subgroup N of A_4 of index 3 is behind the solution of the quartic by radicals, while the absence of such normal subgroups in A_5 explains why there is no formula for the quintic (Galois theory).

Definition 12.2.18 (Center)

$Z(G) = \{z \in G : zg = gz \text{ for all } g \in G\}$ is the **center** of G.

Exercise 12.2.19

(a) $Z(G) \trianglelefteq G$.
(b) If $\varphi \in \mathrm{Aut}(G)$, then $\varphi(Z(G)) = Z(G)$.

Solution. (a) $Z(G)$ is a subgroup, and $g^{-1}zg = z$ for $z \in Z(G)$.

(b) If z commutes with all g, then $\varphi(z)$ commutes with all $\varphi(g)$, i.e. with all of G; apply the same to φ^{-1}. □

Exercise 12.2.20

(a) $Z(\mathrm{GL}_n(F)) = \{cI : c \in F^\times\}$.
(b) $Z(\mathrm{SL}_n(F)) = \{cI : c \in F,\ c^n = 1\}$.

Solution. (a) Scalars are central. If A commutes with $I + E_{ij}$ $(i \neq j)$, it commutes with E_{ij}; $AE_{ij} = E_{ij}A$ forces $A_{ki} = 0$ for $k \neq i$, $A_{jk} = 0$ for $k \neq j$, and $A_{ii} = A_{jj}$. So A is scalar.

(b) The same argument works, since $I + E_{ij} \in \mathrm{SL}_n(F)$; and $cI \in \mathrm{SL}_n(F)$ iff $c^n = 1$. □

Definition 12.2.21 (Projective linear groups)

$\mathbf{PGL}_n(F)$ $= \mathrm{GL}_n(F)/Z(\mathrm{GL}_n(F))$ and **$\mathbf{PSL}_n(F)$** $= \mathrm{SL}_n(F)/Z(\mathrm{SL}_n(F))$; similarly **PSO**, **PSU**, ...

H. Weyl called the groups GL, **PGL**, SL, **PSL**, **O**, **SO**, **PSO**, **U**, **SU**, **PSU** (and **Sp**, **PSp** of Chapter 13) the **classical groups**; introducing them is one aim of this book.

12.3 Quotient spaces

Motivation. For a subspace $W \leq V$ the coset group V/W always admits scalar multiplication, so it is a vector space of dimension $\dim V - \dim W$. This is the basis of induction on dimension.

For $W \leq V$ (any dimension), $(W, +)$ is a normal subgroup of the abelian group $(V, +)$, so V/W is an abelian group with $\bar{u} + \bar{v} = \overline{u+v}$; the cosets are $\bar{v} = v + W$. Define $c\bar{v} = \overline{cv}$. This is well defined: if $\bar{v} = \bar{u}$, then $v - u \in W$, so $cv - cu = c(v-u) \in W$.

Proposition 12.3.1

For $W \leq V$, V/W with $\bar{u} + \bar{v} = \overline{u+v}$ and $c\bar{v} = \overline{cv}$ is a vector space, the **quotient space** of V modulo W.

Proof. The operations are well defined, and each axiom for V/W follows from the same axiom in V by putting bars on; the zero vector is $\bar{0}$ and $-\bar{v} = \overline{-v}$. ■

Example 12.3.2

$V/0 \approx V$ and $V/V \approx 0$.

The natural projection $\pi : V \to V/W$, $\pi(v) = \bar{v}$, is now F-linear.

Exercise 12.3.3

For $W \leq V$: (a) $\pi : V \to V/W$ is a linear epimorphism;
(b) $\ker \pi = W$;
(c) for the inclusion $\iota : W \to V$, $\pi \circ \iota = 0$.

Solution. $\pi(cv + u) = \overline{cv+u} = c\bar{v} + \bar{u}$; onto by definition; $\bar{v} = \bar{0} \iff v \in W$. □

Remark 12.3.4

A heuristic: if $|F| = q < \infty$, $\dim V = n$, $\dim W = m$, then $|V| = q^n$, $|W| = q^m$, and by Lagrange $|V/W| = q^{n-m}$, suggesting $\dim V/W = n - m$.

Theorem 12.3.5

If V is finite-dimensional and $W \leq V$, then $\dim V = \dim W + \dim(V/W)$.

Proof. Apply the dimension theorem to $\pi : V \to V/W$ ($\ker \pi = W$, $\operatorname{im} \pi = V/W$). ■

Exercise 12.3.6

If a basis $\{v_1, \dots, v_m\}$ of W is extended to a basis $\{v_1, \dots, v_n\}$ of V, then $\{\overline{v_{m+1}}, \dots, \overline{v_n}\}$ is a basis of V/W.

Solution. They span: $\bar{v} = \overline{\sum a_i v_i} = \sum_{i>m} a_i \bar{v}_i$ since $\bar{v}_i = \bar{0}$ for $i \le m$. They are independent: if $\sum_{i>m} a_i \bar{v}_i = \bar{0}$, then $\sum_{i>m} a_i v_i \in W = \langle v_1, \dots, v_m \rangle$, so all $a_i = 0$. □

Observation 12.3.7

Every subspace of F^n is the solution space of $AX = 0$ for some $A \in \mathfrak{M}_{m,n}(F)$.

Proof. For $W \le F^n$ it suffices to find a linear $L : F^n \to U$ (finite-dimensional U) with $\ker L = W$; take $L = \pi : F^n \to F^n/W$, and A its matrix in some bases. ■

12.4 Isomorphism theorems

Motivation. A homomorphism identifies elements with the same image. Dividing out its kernel turns it into an isomorphism onto its image (First Isomorphism Theorem), and $K/(H \cap K) \approx HK/H$ (Second). Together they classify cyclic groups and give many quotient computations.

Observation 12.4.1

If $\varphi : G \to H$ is a homomorphism and $\varphi(g) = h$, then $\varphi^{-1}(h) = g \ker \varphi$.

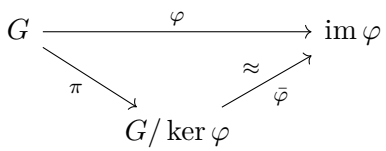
Proof. $x \in \varphi^{-1}(h) \iff \varphi(x) = \varphi(g) \iff g^{-1}x \in \ker \varphi \iff x \in g \ker \varphi$. ■

So the fibres of φ are exactly the cosets of $\ker \varphi$, which is normal; renaming each coset $\bar{g}$ as $\varphi(g)$ is a bijection $G/\ker \varphi \to \operatorname{im} \varphi$.

Theorem 12.4.2 (First isomorphism theorem for groups)

For a homomorphism $\varphi : G \to H$, $\bar{\varphi} : G/\ker \varphi \to \operatorname{im} \varphi$, $\bar{\varphi}(\bar{g}) = \varphi(g)$, is a well-defined isomorphism: $G/\ker \varphi \approx \operatorname{im} \varphi$.

Proof. Well defined: if $\bar{g} = \bar{g}'$, then $g'^{-1}g \in \ker \varphi$, so $\varphi(g) = \varphi(g')$. Homomorphism: $\bar{\varphi}(\bar{g}\bar{g}') = \varphi(gg') = \varphi(g)\varphi(g')$. Onto: clear. Injective: if $\bar{\varphi}(\bar{g}) = 1$, then $g \in \ker \varphi$, so $\bar{g} = \bar{1}$ (Observation 11.6.7). ■



Remark 12.4.3

(a) Writing $\bar{G} = G/\ker \varphi$, $\bar{\varphi}\bar{g} = \varphi g$. (b) Equivalently, $\psi : \operatorname{im} \varphi \to G/\ker \varphi$, $\psi(\varphi(g)) = \bar{g}$, is an isomorphism; its well-definedness ($\varphi(g) = \varphi(g') \Rightarrow \bar{g} = \bar{g}'$) is the injectivity of $\bar{\varphi}$.

Example 12.4.4

sgn : $S_n \to \mu_2$ ($n \geq 2$) is onto with kernel A_n; so $A_n \trianglelefteq S_n$, $S_n/A_n \approx \mu_2$, $[S_n : A_n] = 2$, $S_n = A_n \amalg \tau A_n$ ($\tau \notin A_n$), and $|A_n| = n!/2$.

Example 12.4.5

$\det : \mathrm{GL}_n(F) \to F^\times$ is onto with kernel $\mathrm{SL}_n(F)$; so $\mathrm{SL}_n(F) \trianglelefteq \mathrm{GL}_n(F)$ and $\mathrm{GL}_n(F)/\mathrm{SL}_n(F) \approx F^\times$.

Example 12.4.6

(a) $\det : \mathbf{O}(n) \to \mu_2$ is onto with kernel $\mathbf{SO}(n)$; so $\mathbf{SO}(n) \trianglelefteq \mathbf{O}(n)$, $\mathbf{O}(n)/\mathbf{SO}(n) \approx \mu_2$, and $\mathbf{O}(n) = \mathbf{SO}(n) \amalg S \cdot \mathbf{SO}(n)$ ($S \notin \mathbf{SO}(n)$; Exercise 9.3.11).
(b) $\det : \mathbf{U}(n) \to S^1$ is onto (e.g. $\mathrm{diag}(z, 1, \ldots, 1)$) with kernel $\mathbf{SU}(n)$; so $\mathbf{U}(n)/\mathbf{SU}(n) \approx S^1 \approx \mathbf{SO}(2)$.

Exercise 12.4.7

(a) For $A, B \in \mathrm{GL}_n(F)$, $\bar{A} = \bar{B}$ in $\mathrm{GL}_n(F)/\mathrm{SL}_n(F)$ iff $\det A = \det B$. (b) Describe the coset decomposition of $\mathrm{GL}_n(F)$ modulo $\mathrm{SL}_n(F)$. (c) Describe that of $\mathbf{U}(n)$ modulo $\mathbf{SU}(n)$.

Solution. (a) $\bar{A} = \bar{B} \iff B^{-1}A \in \mathrm{SL}_n(F) \iff \det A = \det B$. (b) $\mathrm{GL}_n(F) = \coprod_{c \in F^\times} \mathrm{diag}(c, 1, \ldots, 1)\,\mathrm{SL}_n(F)$: one coset for each value of the determinant. (c) $\mathbf{U}(n) = \coprod_{z \in S^1} \mathrm{diag}(z, 1, \ldots, 1)\mathbf{SU}(n)$. □

Exercise 12.4.8

$\rho : (\mathbb{R}, +) \to \mathbf{SO}(2)$, $\rho(\theta) = R_\theta$, is a homomorphism, and $\mathbb{R}/2\pi\mathbb{Z} \approx \mathbf{SO}(2) \approx S^1$.

Solution. $R_\theta R_\eta = R_{\theta+\eta}$; ρ is onto (Theorem 9.5.2) with kernel $2\pi\mathbb{Z}$; apply Theorem 12.4.2 and Example 11.9.19. □

Example 12.4.9

For $x \in G$, $\gamma_x : \mathbb{Z} \to G$, $\gamma_x(a) = x^a$, has $\ker \gamma_x = n\mathbb{Z}$ for some $n \geq 0$ (Theorem 11.4.2) and image $\langle x \rangle$, so $\langle x \rangle \approx \mathbb{Z}/n\mathbb{Z} = \mathbb{Z}_n$ (with $\mathbb{Z}_0 = \mathbb{Z}$, $\mathbb{Z}_1 = 0$).

Theorem 12.4.10

Up to isomorphism the cyclic groups are exactly the $\mathbb{Z}_n$, $n \geq 0$.

Proof. Example 12.4.9 with $G = \langle x \rangle$; and $\mathbb{Z}_n$ is cyclic. ■

Exercise 12.4.11

For $\mathrm{Int} : G \to \mathrm{Aut}(G)$, $g \mapsto \mathrm{Int}(g)$: (a) Int is a homomorphism;
(b) $\mathrm{Int}(G) \trianglelefteq \mathrm{Aut}(G)$;
(c) $\ker(\mathrm{Int}) = Z(G)$, so $\mathrm{Int}(G) \approx G/Z(G)$.

Solution. (a) $\mathrm{Int}(gh)(x) = ghxh^{-1}g^{-1} = \mathrm{Int}(g)\,\mathrm{Int}(h)(x)$ (this is why $\mathrm{Int}(g)(x) = gxg^{-1}$ rather than $g^{-1}xg$). (b) For $\varphi \in \mathrm{Aut}(G)$: $\varphi \circ \mathrm{Int}(g) \circ \varphi^{-1}(x) = \varphi(g\varphi^{-1}(x)g^{-1}) = \varphi(g)x\varphi(g)^{-1}$, i.e. $= \mathrm{Int}(\varphi(g))$. (c) $\mathrm{Int}(g) = id \iff gx = xg$ for all x; apply Theorem 12.4.2. □

Exercise 12.4.12

Define $\psi : \mathrm{SL}_2(F) \to \mathrm{GL}_3(F)$ by

$$\psi\begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} a^2 & -2ac & -c^2 \\ -ab & ad+bc & cd \\ -b^2 & 2bd & d^2 \end{pmatrix}.$$

(a) $\mathrm{im}\,\psi \leq \mathrm{SL}_3(F)$.
(b) ψ is a homomorphism with $\ker\psi = \{\pm I\} = Z(\mathrm{SL}_2(F))$.
(c) Hence $\mathbf{PSL}_2(F) \approx \mathrm{im}\,\psi$ is a matrix group.

Solution. [Supplement] $\psi(A)$ records how A acts on degree-2 polynomials in two variables, which is why it is multiplicative. Concretely, multiplying out (we checked it symbolically) gives $\psi(AB) = \psi(A)\psi(B)$ for all 2×2 matrices A, B, and $\det\psi(A) = (ad-bc)^3$. (a) For $A \in \mathrm{SL}_2(F)$, $\det\psi(A) = 1^3 = 1$. (b) $\psi(A) = I$ forces $c = b = 0$ (off-diagonal entries $-c^2 = -b^2 = 0$) and $a^2 = d^2 = ad = 1$, so $a = d = \pm 1$; conversely $\psi(\pm I) = I$. And $Z(\mathrm{SL}_2(F)) = \{\pm I\}$ (Exercise 12.2.20, $c^2 = 1$). (c) First isomorphism theorem. □

Exercise 12.4.13

Is there an epimorphism $S_3 \to \mathbb{Z}_3$?

Solution. No. Its kernel would be a normal subgroup of order 2 (First Isomorphism Theorem), but the subgroups of order 2 of S_3 are not normal (Example 12.2.1). □

Theorem 12.4.14 (First isomorphism theorem for vector spaces)

For a linear map $L : V \to W$, $\bar{L} : V/\ker L \to \mathrm{im}\,L$, $\bar{L}\bar{v} = Lv$, is a well-defined vector space isomorphism.

Proof. By Theorem 12.4.2 for $L : (V,+) \to (W,+)$, $\bar{L}$ is a well-defined group isomorphism; and $\bar{L}(c\bar{v}) = \bar{L}(\overline{cv}) = L(cv) = cL(v) = c\bar{L}(\bar{v})$. ■

Remark 12.4.15

By Theorem 12.4.14, the dimension theorem and Theorem 12.3.5 are equivalent statements.

Exercise 12.4.16

For $\mathrm{tr} : \mathfrak{M}_{n,n}(F) \to F$: (a) $\ker\mathrm{tr} = \mathfrak{sl}_n(F)$;
(b) $\mathfrak{M}_{n,n}(F)/\mathfrak{sl}_n(F) \approx F$, so $\dim\mathfrak{sl}_n(F) = n^2 - 1$.

Solution. tr is linear and onto ($\mathrm{tr}(cE_{11}) = c$); $\mathfrak{sl}_n(F)$ is by definition the trace-zero matrices; apply Theorem 12.4.14. □

Exercise 12.4.17

For $T \in \mathfrak{L}(V,V)$ and a T-cyclic space $V = F[t]v$, $\varphi : F[t] \to V$, $\varphi(f) = f(T)v$: (a) φ is linear with $\ker\varphi = \mathcal{I}_T$ (the annihilator ideal of v);
(b) $V \approx F[t]/\mathcal{I}_T$.

Solution. φ is linear and onto by definition of $F[t]v$; its kernel is $\{f : f(T)v = 0\}$; apply Theorem 12.4.14. □

Observation 12.4.18

If $H \trianglelefteq G$ and $K \leq G$, then $HK \leq G$.

Proof. $1 \in HK$. For $h,h' \in H$, $k,k' \in K$: $(hk)(h'k') = h(kh'k^{-1})kk' \in HK$ and $(hk)^{-1} = k^{-1}h^{-1} = (k^{-1}h^{-1}k)k^{-1} \in HK$. ■

Theorem 12.4.19 (Second isomorphism theorem for groups)

If $H \trianglelefteq G$ and $K \leq G$, then $\bar{\jmath} : K/(H \cap K) \to HK/H$, $\bar{\jmath}(k(H \cap K)) = kH$, is a well-defined isomorphism.

Proof. By Observation 12.4.18 $HK \leq G$, and clearly $H \trianglelefteq HK$, $H \cap K \trianglelefteq K$.

(i) Well defined: if $k'^{-1}k \in H \cap K$, then $k'^{-1}k \in H$.

(ii) Injective: if $k'^{-1}k \in H$, then (as $k,k' \in K$) $k'^{-1}k \in H \cap K$.

(iii) Homomorphism: $\bar{\jmath}(\bar{k}\bar{k}') = \overline{kk'} = \bar{k}\,\bar{k}'$.

(iv) Onto: for $h \in H$, $k \in K$, $\overline{hk} = \bar{h}\,\bar{k} = \bar{1}\bar{k} = \bar{k}$ in HK/H (H normal). ■

Second proof. $j = \pi' \circ \iota : K \to HK \to HK/H$ is an epimorphism by (iv), with $\ker j = H \cap K$; apply Theorem 12.4.2. ■

Corollary 12.4.20

If G is finite, $H \trianglelefteq G$, $K \leq G$, then $|HK| \cdot |H \cap K| = |H| \cdot |K|$.

Proof. Lagrange and Theorem 12.4.19: $\frac{|K|}{|H \cap K|} = \frac{|HK|}{|H|}$. ■

Theorem 12.4.21 (Second isomorphism theorem for vector spaces)

For $U,W \leq V$, $\bar{\jmath} : W/(U \cap W) \to (U+W)/U$, $\bar{\jmath}(\bar{w}) = \bar{w}$, is a well-defined vector space isomorphism.

Proof. Theorem 12.4.19 for $(V,+)$ gives a group isomorphism; it is F-linear since $\bar{\jmath}(c\bar{w}) = \overline{cw} = c\bar{w}$. ■

Corollary 12.4.22

For finite-dimensional V and $U,W \leq V$: $\dim(U+W) + \dim(U \cap W) = \dim U + \dim W$.

Proof. By Theorems 12.4.21 and 12.3.5, $\dim W - \dim(U \cap W) = \dim(U+W) - \dim U$. ■

12.5 Triangularization II

Motivation. An invariant subspace W splits an operator into $T|_W$ and the induced operator $\bar{T}$ on V/W, with block-triangular matrix. Induction on $\dim V$ via quotients then proves triangularization over $\mathbb{C}$ and simultaneous triangularization of commuting operators.

Throughout, spaces are finite-dimensional. If W is T-invariant with basis $\mathfrak{C} = \{v_1, \dots, v_m\}$ extended to a basis $\mathfrak{B} = \{v_1, \dots, v_n\}$ of V, then $[T]_{\mathfrak{B}}^{\mathfrak{B}} = \left(\begin{smallmatrix} [T|_W]_{\mathfrak{C}}^{\mathfrak{C}} & * \\ 0 & ? \end{smallmatrix}\right)$. The block "?" is explained by $\bar{V} = V/W$ with basis $\bar{\mathfrak{B}} = \{\bar{v}_{m+1}, \dots, \bar{v}_n\}$ (Exercise 12.3.6) and $\bar{T}\bar{v} = \overline{Tv}$.

Observation 12.5.1

If $W \leq V$, $W' \leq V'$ and $L : V \to V'$ is linear with $LW \leq W'$, then $\bar{L} : V/W \to V'/W'$, $\bar{L}\bar{v} = \overline{Lv}$, is a well-defined linear map (L **induces** $\bar{L}$).

Proof. If $v - u \in W$, then $Lv - Lu = L(v - u) \in W'$; linearity is inherited. (Conversely, well-definedness forces $LW \leq W'$: take $u = 0$.) ■

$$\begin{array}{ccc} V & \xrightarrow{\;L\;} & V' \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi} \\ V/W & \xrightarrow[\bar{L}]{} & V'/W' \end{array}$$

Exercise 12.5.2

State and prove Observation 12.5.1 for groups.

Solution. If $N \trianglelefteq G$, $N' \trianglelefteq G'$ and $\varphi : G \to G'$ is a homomorphism with $\varphi(N) \leq N'$, then $\bar{\varphi}(\bar{g}) = \overline{\varphi g}$ is a well-defined homomorphism $G/N \to G'/N'$: if $g'^{-1}g \in N$, then $\varphi(g')^{-1}\varphi(g) = \varphi(g'^{-1}g) \in N'$. □

Exercise 12.5.3

The well-definedness of $\bar{L}$ in Theorem 12.4.14 and of $\bar{\jmath}$ in Theorem 12.4.21 are special cases of Observation 12.5.1.

Solution. $\bar{L}$ is induced by $L : V \to \operatorname{im} L$ with $L(\ker L) \leq 0$; $\bar{\jmath}$ is induced by the inclusion $W \to U+W$ with $U \cap W \leq U$. □

Lemma 12.5.4

Let W be T-invariant, $\bar{T} : \bar{V} \to \bar{V}$ as above, $\mathfrak{C} = \{v_1, \dots, v_m\}$ a basis of W, $\mathfrak{B} = \{v_1, \dots, v_n\}$ a basis of V, $\bar{\mathfrak{B}} = \{\bar{v}_{m+1}, \dots, \bar{v}_n\}$. Then

$$[T]_{\mathfrak{B}}^{\mathfrak{B}} = \begin{pmatrix} [T|_W]_{\mathfrak{C}}^{\mathfrak{C}} & * \\ 0 & [\bar{T}]_{\bar{\mathfrak{B}}}^{\bar{\mathfrak{B}}} \end{pmatrix}.$$

Proof. Let column $m + j$ of $[T]_{\mathfrak{B}}^{\mathfrak{B}}$ be $(a_{1,m+j}, \dots, a_{n,m+j})^t$: $Tv_{m+j} = \sum_i a_{i,m+j}v_i$. Applying bars and using $\bar{v}_1 = \cdots = \bar{v}_m = \bar{0}$, $\bar{T}\bar{v}_{m+j} = \sum_{i>m} a_{i,m+j}\bar{v}_i$. So the j-th column of $[\bar{T}]$ is the lower part of column $m + j$. The first m columns are as stated because $TW \leq W$. ■

Exercise 12.5.5

In Lemma 12.5.4: (a) $\operatorname{tr}T = \operatorname{tr}(T|_W) + \operatorname{tr}\bar{T}$;
(b) $\det T = \det(T|_W)\det\bar{T}$;
(c) $\phi_T(t) = \phi_{T|_W}(t)\phi_{\bar{T}}(t)$.

Solution. Read them off the block-triangular matrix (the determinant of a block-triangular matrix is the product of the determinants of the diagonal blocks; apply this to $tI - [T]$ for (c)). □

Theorem 12.5.6 (Triangularization)

If $F = \mathbb{C}$ and $T \in \mathfrak{L}(V, V)$, then V has a basis $\mathfrak{B}$ with $[T]_{\mathfrak{B}}^{\mathfrak{B}}$ upper triangular.

Second proof. Induction on $n = \dim V$; $n = 1$ is trivial. Over $\mathbb{C}$, T has an eigenvector v, $Tv = \lambda v$; $W = \langle v\rangle$ is T-invariant. With $\bar{V} = V/W$, Lemma 12.5.4 gives $[T]_{\mathfrak{B}}^{\mathfrak{B}} = \left(\begin{smallmatrix}\lambda & * \\ 0 & [\bar{T}]\end{smallmatrix}\right)$, and $\dim\bar{V} = n-1$. By induction $\bar{V}$ has a basis $\bar{\mathfrak{D}} = \{\bar{u}_2, \ldots, \bar{u}_n\}$ with $[\bar{T}]_{\bar{\mathfrak{D}}}^{\bar{\mathfrak{D}}}$ upper triangular. Then $\mathfrak{D} = \{v, u_2, \ldots, u_n\}$ is a basis of V: if $cv + \sum c_i u_i = 0$, applying π gives $\sum c_i\bar{u}_i = 0$, so all $c_i = 0$, then $c = 0$. By Lemma 12.5.4, $[T]_{\mathfrak{D}}^{\mathfrak{D}} = \left(\begin{smallmatrix}\lambda & * \\ 0 & [\bar{T}]_{\bar{\mathfrak{D}}}^{\bar{\mathfrak{D}}}\end{smallmatrix}\right)$ is upper triangular. ■

Example 12.5.7

Let G be finite and $\varphi : G \to \mathrm{GL}_n(\mathbb{C})$ a homomorphism, $\chi = \operatorname{tr}\circ\varphi$ (the **character** of φ). For $g \in G$, triangularize $\varphi(g)$ with diagonal $\lambda_1, \ldots, \lambda_n$. Then $I = \varphi(g^{|g|}) = \varphi(g)^{|g|}$ has diagonal $\lambda_i^{|g|}$, so $\lambda_i^{|g|} = 1$ and $|\lambda_i| = 1$. Since $\varphi(g^{-1}) = \varphi(g)^{-1}$ is triangular with diagonal $\lambda_i^{-1} = \bar{\lambda}_i$,

$$\chi(g^{-1}) = \sum \lambda_i^{-1} = \sum \bar{\lambda}_i = \overline{\chi(g)}.$$

Exercise 12.5.8

In Example 12.5.7, if g is conjugate to h, then $\chi(g) = \chi(h)$.

Solution. $h = x^{-1}gx$ gives $\varphi(h) = \varphi(x)^{-1}\varphi(g)\varphi(x)$, and similar matrices have equal traces. □

Definition 12.5.9 (Simultaneous triangularization)

$T, S \in \mathfrak{L}(V, V)$ are **simultaneously triangularizable** if $[T]_{\mathfrak{B}}^{\mathfrak{B}}$ and $[S]_{\mathfrak{B}}^{\mathfrak{B}}$ are upper triangular for one basis $\mathfrak{B}$ (likewise for any family). Then the first vector of $\mathfrak{B}$ is a common eigenvector.

Proposition 12.5.10

If $F = \mathbb{C}$ and $TS = ST$, then T and S have a common eigenvector.

Proof. Take an eigenvalue λ of T and its eigenspace E_λ^T. It is S-invariant: if $Tv = \lambda v$, then $T(Sv) = S(Tv) = \lambda Sv$. An eigenvector of $S|_{E_\lambda^T}$ (which exists over $\mathbb{C}$) is a common eigenvector. ■

Proposition 12.5.11

If $F = \mathbb{C}$ and $TS = ST$, then T, S are simultaneously triangularizable.

Proof. Take a common eigenvector v (Proposition 12.5.10) and $\bar{V} = V/\langle v\rangle$. $\langle v\rangle$ is invariant under both, so $\bar{T}, \bar{S}$ are defined, and $\bar{T}\bar{S} = \overline{TS} = \overline{ST} = \bar{S}\bar{T}$. By induction on dimension they are

simultaneously triangularizable on $\bar{V}$, and the lifting argument of Theorem 12.5.6 applies to both at once. ■

Exercise 12.5.12

If $\{T_i\}_{i\in I} \subseteq \mathfrak{L}(V,V)$ $(F=\mathbb{C})$ pairwise commute, they are simultaneously triangularizable.

Solution.

Step 1 (a common eigenvector). Choose a nonzero subspace U invariant under all T_i of minimal dimension (e.g. V at first). If some $T_j|_U$ is not a scalar, pick an eigenvalue λ of $T_j|_U$; then $\{u \in U : T_j u = \lambda u\}$ is a smaller nonzero subspace invariant under all T_i (commutativity), contradicting minimality. So every T_i acts on U as a scalar, and any $0 \neq v \in U$ is a common eigenvector.

Step 2 (induction). Pass to $V/\langle v\rangle$, where the induced operators still commute, and argue as in Proposition 12.5.11. □